Chapter 3
Impact of the Loss of *Titanic* on Maritime Safety and International Law and Policy



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Abstract The tragic loss of *Titanic* was a catalyst for improvements to maritime safety law and policy including ensuring enough lifeboats for passengers and crew, safety drills, wireless communication improvements, ice patrols and improvements in ship design. The international cooperation led to the first treaty on maritime safety and navigation the Safety of Life at Sea Convention (SOLAS) and eventually the International Maritime Organization (IMO) that continues to work on improving safety in navigation and preventing pollution of the environment from ships.

Keywords Maritime Safety · Safety of Life at Sea Convention (SOLAS) · International Maritime Organization (IMO) · Maritime disaster · International law · Regulation

The tragic loss of the RMS *Titanic* in 1912 can be seen as a catalytic moment that transformed maritime safety worldwide. The disaster exposed fatal gaps in ship design, navigational practices, lifeboat provision, wireless communication, emergency procedure, and rescue response. This chapter explores the impacts of both the loss of *Titanic* and other significant maritime disasters and the lessons that can continue to be learned from them. It also seeks to outline what has evolved into an increasingly complex international regulatory landscape, one designed to ensure the safety of people and of the environment.

Safety is often only reviewed at an international level following a major disaster. The tragic loss of over 1500 lives with the sinking of *Titanic* ultimately led to international collaboration to improve safety at sea. But why this event? Would the question of safety at sea have been tackled if *Titanic* had not hit an iceberg? Surely, some international regulation would become necessary sooner or later (Bedford & Hackett, 1996: 45). The global nature of the industry, with vessel ownership, registration, shipbuilding, and crews divided across many nations, meant that there were limited consistent mechanisms to ensure safety at sea and the prevention of damage to the environment. Perhaps without *Titanic*, the loss of the Canadian-owned *Empress of Ireland* on 29 May 1914 would have been the trigger.

A total of 1021 lives were lost with its sinking in the Saint Lawrence River following a nighttime collision with the Norwegian coal freighter *Storstad*. The

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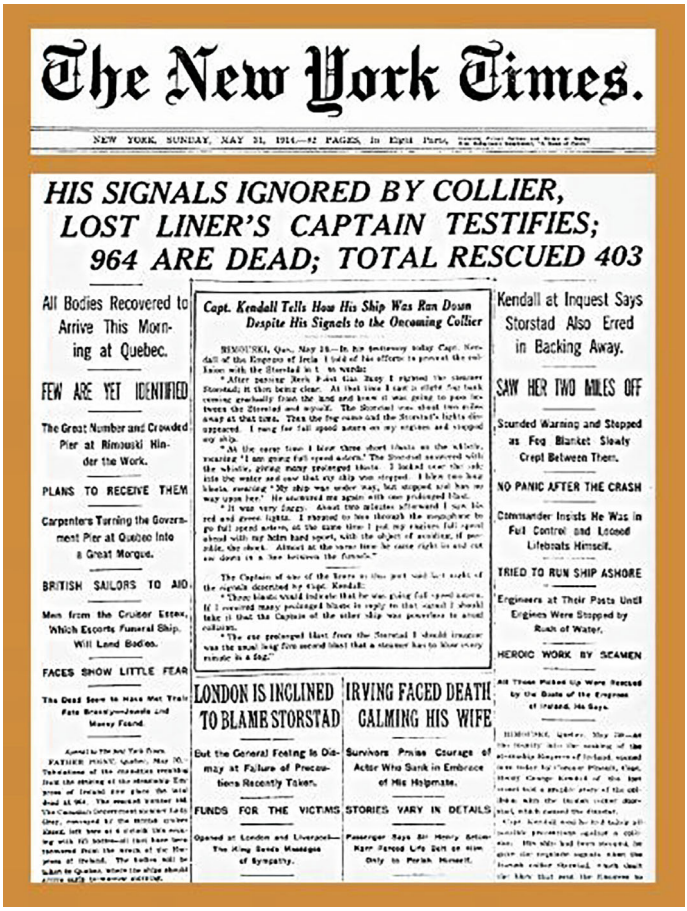
impact with *Storstad*'s ice-breaking bow ripped through the *Empress of Ireland*'s starboard side, soon after which its boilers exploded. The ship sank in just 17 min with only 446 passengers rescued, compared to 705 from *Titanic* (Butler, 2011: 239; also see Zeni, 1998). The Canadian government launched an inquiry into the disaster, which was described by *The Illustrated London News* as "The Greatest Disaster Since the 'Titanic' was struck by ice" (Illustrated London News, 1914).

3.1 International Inquiry Into the Sinking

The sinking of the *Titanic* prompted an investigation in the US and in Britain. The US Congressional hearings resulted in a report and domestic safety measures (Levin Center, n.d.). The British Government called for an international inquiry into the circumstances of the casualty. It concluded that the excessive speed at which the ship was being navigated caused the collision with the iceberg (British Wreck Commissioner, 1912). The comments by Sir Alfred Chalmers, the Board of Trade representative at the time of the loss of *Titanic,* that there was nothing new to be learnt from the accident because "it was an extraordinary incident," were quickly contradicted by the recommendations and actions that followed the inquiry. The findings of the inquiry prompted significant changes in maritime safety practices, including: the provision of lifeboats based on passenger numbers rather than on tonnage, lifeboat and other drills to be carried out after sailing, installation of wireless telegraphy, a 24-h radio watch enabled by having a sufficient number of trained operators to ensure a continuous service, suggestions for improved watertight sub-division, the International Ice Patrol, lookouts, and the first International Convention for the Safety of Life at Sea (Bedford & Hackett, 1996: 58). It is fundamental that we continue to study maritime casualties in this way, to find and revive mechanisms to ensure safety at sea, and that we remember and act upon these lessons (Fig. 3.1).

3.2 Lessons Learned from the Loss of *Britannic*

Immediate lessons were evident in changes made to another vessel being built at Harland and Wolff – *Britannic*. Because of the *Titanic* disaster, it was fitted with higher and stronger bulkheads that reached all the way up to B deck which was 40 feet above the deepest load line. *Titanic*'s bulkheads had reached only to E deck, just 11 feet above the load line (Britton, 2017). Unlike its two sisters (*Olympic* and *Titanic*), which could have any two adjoining watertight compartments flooded and still survive, it could theoretically stay afloat with any six compartments flooded (one more than the number of compartments breached on *Titanic*). It also had new improved lifeboat davits, which were powered by electric motors, each could launch up to six lifeboats over the side of the ship, regardless of listing (Lawrence, 2004: 213–214).



The New York Times.

NEW YORK, SUNDAY, MAY 21, 1914.—82 PAGES. In Eight Parts.

HIS SIGNALS IGNORED BY COLLIER, LOST LINER'S CAPTAIN TESTIFIES; 964 ARE DEAD; TOTAL RESCUED 403

All Bodies Recovered to Arrive This Morning at Quebec.

FEW ARE YET IDENTIFIED

The Great Number and Crowded Pier at Rimouski Hinder the Work.

PLANS TO RECEIVE THEM

Carpenters Turning the Government Pier at Quebec Into a Great Morgue.

BRITISH SAILORS TO AID

Men from the Cruiser Essex, Which Escorts Funeral Ship, Will Land Bodies.

FACES SHOW LITTLE FEAR

The Dead Seem to Have Met Their Fate Bravely—Jewels and Money Found.

Capt. Kendall Tells How His Ship Was Run Down Despite His Signals to the Oncoming Collier

Kendall at Inquest Says Storstad Also Erred in Backing Away.

SAW HER TWO MILES OFF

Sounded Warning and Stopped as Fog Blanket Slowly Crept Between Them.

NO PANIC AFTER THE CRASH

Commander Insists He Was in Full Control and Loosed Lifeboats Himself.

TRIED TO RUN SHIP ASHORE

Engineers at Their Posts Until Engines Were Stopped by Rush of Water.

HEROIC WORK BY SEAMEN

All Those Picked Up Were Rescued by the Boats of the Empress of Ireland, He Says.

LONDON IS INCLINED TO BLAME STORSTAD

But the General Feeling Is Dismay at Failure of Precautions Recently Taken.

FUNDS FOR THE VICTIMS

Opened at London and Liverpool—The King Sends Messages of Sympathy.

IRVING FACED DEATH CALMING HIS WIFE

Survivors Praise Courage of Actor Who Sank in Embrace of His Helpmate.

STORIES VARY IN DETAILS

Passenger Says Sir Henry Actually Kept Fenced Life Belt on Him—Only to Perish Himself.

Fig. 3.1 A USCG C-130 flight during International Ice Patrol operations (USCG, public domain)

The test of these modifications came in November 1916, when *Britannic*, while serving as a hospital ship during World War I, struck an underwater mine and sank off the coast of Kea, Greece. Although listing, the new equipment successfully enabled the launch of 35 lifeboats, and it made distress calls. Unfortunately, while ploughing on under its own steam in an attempt to beach the vessel, lifeboats were drawn into the huge propellers "churning and mincing up everything near them—men, boats, and everything were just one ghastly whirl" (Lawrence, 2004: 410–412). Twenty-eight lives were lost, and sadly, all those who died had been in the lifeboats.

Britannic sank in just 50 min but the rapid launch of the lifeboats, the warmer water conditions of the Aegean Sea, and faster rescue by nearby vessels meant that a total of 1036 people were saved. In 2003, divers discovered several reasons why *Britannic* had sunk so fast. Inside the wreck they found the watertight doors between the engine room compartments were not closed. Along the lower decks, numerous

portholes were left open by medical staff for ventilation, which was against regulations. Additionally, many of the rivets in the hull had popped out. "This is thought to have been because they were cold riveted (drilled but not welded). In salt water, rivets and plates made of different metals are much more likely to corrode and separate unless they are hot riveted (welded together)" (Lawrence, 2004: 412–413). The maritime industry changes made in the aftermath of *Titanic*'s sinking had a positive impact on lifesaving appliances and communication. Yet there were still lessons to be learned and a need to ensure that safety regulations could be applied more widely. International agreement and enforcement was needed.

3.3 International Convention for the Safety of Life at Sea

In 1914, maritime nations gathered in London adopted the International Convention for the Safety of Life at Sea (SOLAS Convention), considering lessons learned from *Titanic*. This was the first international convention aimed at improving maritime safety and preventing similar tragedies in the future. As the First World War prevented the implementation of the 1914 Convention, a the second version, an outcome of the 1929 Convention, eventually came into force in 1933. Since then, the SOLAS Convention has been revised to cover an increasing range of developments (with a third version in 1948, and a fourth in 1960). The Convention in force today, sometimes referred to as SOLAS, 1974, as amended, covers 15 chapters. These include: certification, casualty investigation and Port State Control; ship design and construction; lifesaving; distress and safety communications (radio and satellite); safety of navigation; carriage of cargoes (loading and securing cargoes, including containers); carriage of dangerous goods (International Maritime Dangerous Good Code); safety of nuclear fueled ships; safety management (International Safety Management (ISM) Code); Safety of high- speed craft; special measures for safety (including IMO number scheme) and maritime security for ships and ports; additional safety measures for bulk carriers; IMO Member State Audit Scheme; ships in polar waters; and safety of ships carrying offshore workers (IMO, n.d).

Each amendment and update is carefully crafted and considered, which often takes several years of research and discussion due to the complex nature of governing an international and ever-changing industry. For example. the 1948 SOLAS Convention set out minimum standards for radar and tightened up radio watch regulations. This version included a mandatory continuous 24-h distress watch, something that had been highlighted by the *Titanic* disaster 36 years earlier (at the time, the *Californian* was less than 20 miles away, but the radio officer had gone off duty when the distress messages were sent) (Hind, n.d.). Under SOLAS Chapter IV, every ship while at sea must maintain a continuous watch on the distress and safety frequencies. SOLAS was also intended to enable study of other aspects of ship design and operation, something that the classification societies have been instrumental in.

The ongoing work and innovation of shipowners, shipbuilders, engineers, and classification societies had led to great strides in technology and multiple safety

improvements. Classification societies like Lloyd's Register have specialist Research and Development departments to oversee vessel design and Committees to develop and refine the structural rules that govern ships' construction. "Theirs is a continuous effort to adapt to the ever-changing landscape of maritime technology and environmental concerns, shaping the future of ship design" (Fields, 2023). It is worth emphasizing that *Titanic* was not classed; it met and indeed exceeded the standards laid out by the Board of Trade at the time of her sailing, but this was still not enough to avert disaster.

3.4 Developments in Search and Rescue

Embryonic search and rescue capabilities, open lifeboat design, lack of passenger drills and crew training also played a role in the loss of life being so high. *Titanic* used radio with a limited range of 200 nautical miles. Although the land station at Cape Race, Newfoundland and ships other than *Carpathia* and *Californian* heard the *Titanic* distress call, the airwaves were crackling, and *Titanic*'s position was misinterpreted. In the time that it took for them to be located and rescued, some people had died from hypothermia in the open lifeboats which gave no protection against the cold. Today, under SOLAS, lifeboats must be fully or partially enclosed (with a collapsible roof to fold across). Ships can communicate globally via satellites, and with Emergency Position-Indicating Radio Beacon (EPIRB), and global positioning systems, the position of a ship in distress can be automatically sent much more accurately ("Surviving disaster—The Titanic and SOLAS", n.d.). The 1974 SOLAS Convention obliged signatories to set up search and rescue facilities in coastal waters and the Inter-Governmental Maritime Consultative Organization (which became the International Maritime Organization in 1982) adopted the International Convention on Maritime Search and Rescue (SAR) in 1979.

Since the loss of *Titanic*, the work of SAR organizations worldwide has also grown. The Global Maritime Distress and Safety System (GMDSS) was adopted in 1988 and began to be phased in from 1992. Search and Rescue Satellite-Aided Tracking (SARSAT) and other technologies have also become increasingly important in protecting life and property by providing accurate, timely, and reliable distress alert and location information (NOAA, n.d). In 2025, the International Maritime Rescue Federation (IMRF), in partnership with the safety charity Lloyd's Register Foundation, launched the first-ever Global Maritime Search and Rescue Systems Review in order to understand, evaluate and help strengthen maritime SAR systems worldwide ("Partnership to lead a strategic review of global maritime search and rescue systems", 2025).

Repeatedly, maritime disasters have shown the need for rapid and informed response. Class societies and commercial technical salvage organizations have increasingly developed and offered emergency support for vessels suffering an emergency event, such as structural damage, instability or pollution. One such example is the Lloyd's Register Ship Emergency Response Service (SERS), which has generated

3D computer models loaded with critical information about each vessel. This preparation ensures that when an emergency occurs the situation can be quickly assessed and prompt support offered. Specialist teams of naval architects, former master mariners and marine engineers provide proactive technical support and advice, precisely simulating and rapidly assessing casualties to ensure the safety of vessels and their crews on the high seas. The SERS team has responded to more than 2500 incidents in over 40 years ("Ship Emergency Response Service", n.d.).

The increasing number and size of cruise ships operating in the Arctic in recent years is also increasing safety concerns regarding rescue capacity and response times in the region. Fellows from the University of the Arctic are investigating the risks associated with cruise ship operations in Arctic waters, aiming to enhance safety protocols and minimize environmental impacts (UArctic, 2025). The IMO Polar Code, an international code of safety for ships operating in polar waters covers the full range of design, construction, equipment, operational, training, search and rescue and environmental protection matters relevant to ships operating in the inhospitable waters surrounding the two poles (IMO). As vessel sizes increase, propulsion methods and fuels change, hazardous cargoes are carried, and new navigational routes open up, the challenges of ensuring safety of people and environment continue to necessitate international cooperation.

3.5 International Cooperation for Maritime Safety

Prior to the catalytic impact of the loss of the *Titanic* on international regulation there had been earlier incidences of international maritime cooperation. One such example related to guidance to avoid collision at sea. Now officially called Collision Regulations or COLREGS, varying rules had been in place for hundreds of years. In 1846 regulations were enacted by the UK Parliament as part of the Steam Navigation Act, having been drawn up by Trinity House. A completely new set of international rules developed by the British Board of Trade in consultation with the French Government came into operation in 1863. By the end of 1864 these regulations, known as Articles, had been adopted by over thirty maritime countries including the US and Germany. In 1889, the first International Maritime Conference to consider regulations for preventing collision at sea was held in Washington. In 1897, the resultant regulations were brought into force by several countries, including Britain and the US. International agreement was again reached at a Maritime Conference held in Brussels in 1910, these rules remained in force until 1954 (Cockroft & Lameijer, 2012). Nevertheless, a collision in 1987 saw the deadliest peacetime maritime disaster in history when the *Doña Paz*, an inter-island passenger ferry in the Philippines collided with the oil tanker *Vector* resulting in the loss of over 4300 lives. The cause of the collision is unclear, but the high loss of life was attributed to significant overcrowding on the *Doña Paz,* which was only registered to carry 1518 passengers (Attwood, 2018).

Another early example of national principles being adopted at international level, and a significant precursor to the SOLAS Convention, is the evolution of load line

regulations to prevent overloading. These established uniform principles and rules for the loading of ships on international voyages and played a crucial role in enhancing maritime safety and protecting lives at sea long before the *Titanic*. The calculation and application of freeboard (load lines) became an early and critical predecessor to formal international agreement. As merchant shipping became increasingly competitive in the mid-nineteenth century, there was growing concern over the high loss of life and property from 'coffin ships'—wooden sailing ships overloaded to maximize profits. Despite the 'Lloyd's Rule', introduced by Lloyd's Register in 1835, many transatlantic ships were still overloaded, as the rule remained voluntary. After years of campaigning by UK Member of Parliament Samuel Plimsoll, the Merchant Shipping Act, 1876 made load lines compulsory on all British vessels and on foreign vessels using British ports (Sanger, 2020). This also highlighted the importance of public and political pressure in advocating for safety at sea; something that the *Titanic* disaster again demonstrated the following century. In 1930, the first International Convention on Load Lines was convened to safeguard life and property at sea, and the 1966 Convention established a more comprehensive and adaptable framework, accommodating a wider range of ship types and operational environments (International Convention on Load Lines 1966). Specific data on the number of ships lost due to overloading is not readily available but there are still significant incidents likely attributable to this cause.

3.6 Establishment of the International Maritime Organization

It became clear following the loss of the *Titanic* that a permanent international body should be established to more effectively promote maritime safety. Although several countries proposed this, it was not until after the creation of the United Nations (UN) in 1945 that it became possible. In 1948, the UN Maritime Conference held in Geneva drafted a convention for the Inter-Governmental Maritime Consultative Organization (IMCO) set up with the prime purpose of 'Safe, Secure and Efficient Shipping on Clean Oceans'. The convention was adopted ten years later, and the IMCO was established in London in 1959 with 21 Member Countries. Its first task was to adopt a new version of the International Convention for the Safety of Life at Sea, which it achieved in 1960. IMCO was renamed the International Maritime Organization (IMO) in 1982 and it continues to work on a permanent basis through committees and subcommittees. Inevitably there are constraints upon large international organizations which require consensus among their members; IMO has (as of 2025) 176 Member States and three Associate Members (IMO, n.d.).

A single international organization is necessary in an industry where national flags and national ownership play a much smaller role in regulation, but it is difficult to get a swift consensus, which sometimes results in action that lags behind technology and other developments (Watson, 2015: 233). Given the huge scale of the

challenges facing the industry, international regulation has often struggled to live up to expectations while uniform enforcement has also proved to be difficult. Despite this, IMO has no doubt had a beneficial impact on maritime safety and by the late 1970s, IMCO had largely achieved its initial aim of formulating international treaties and other legislation concerning safety and marine pollution prevention.

On its formation, the IMCO had taken over the management of the first International Convention for the Prevention of Pollution of the Sea by Oil (OILPOL 54) through its Maritime Safety Committee. Although this covered only operational discharges following tank cleaning, it was an increasingly important subject given the growing size of oil tankers. These transported substantial loads of crude oil capable of inflicting severe environmental damage. But it was to take another major disaster to spur further action. In the midst of the environmental movement, the *Torrey Canyon* disaster off the Isles of Scilly in 1967 prompted the parties of the IMCO to develop several international environmental laws regarding pollution from vessels: the Convention on Civil Liability for Oil Pollution Damage (1969) (placing liability on shipowners), the Convention Relating to Intervention on the High Seas (allowing coastal states to act against pollution from ships in international waters), and the Fund Convention (1971) (creating a fund for additional compensation when damages exceed the limits of the Civil Liability Convention). Additionally, it spurred the development of the 1973 International Convention for the Prevention of Pollution from Ships (MARPOL 1973), which became perhaps its most important international treaty on preventing pollution from vessels. The devastating effect of the *Exxon Valdez* oil spill off the coast of Alaska in 1989 led to the introduction of the Oil Pollution Act in 1990 (OPA 90), when the US insisted that, in the future, only double-hulled tankers would be permitted in its waters. Although a 30-year phasing-out period was conceded for existing tankers, this principle was adopted by the IMO in 1994, demonstrating how national disaster response was taken on at international level.

SOLAS conferences continued to be held after the establishment of IMCO; a key role was in reviewing previous decisions in the light of new technological and other developments. The 1960 SOLAS Convention adopted Very High Frequency (VHF) radio for watch-keeping, which in 1981 became compulsory for almost all shipping. In 1977, the IMCO initiated a standard marine navigational vocabulary for voluntary adoption (Watson, 2015: 229). Renamed the IMO Standard Marine Communication Phrases (SMCP) in 2001, this helps to overcome the confusion sometimes facing multi-national crews over terms used in English as the international language of the sea (IMO, 2001).

Due to an increase in the rate of accidents at sea, the IMCO began considering the need for better training in the late 1960s. It finally secured agreement in 1978 for moves to improve worldwide safety and training standards for mariners; countries representing 93% of world tonnage signed up to the International Convention on Standards of Training, Certification, and Watchkeeping for Seafarers (STCW). However, difficulties arose out of the cultural diversity among signatories, and among crews, meaning that the STCW Code was not uniformly applied or properly enforced. This led to its revision in 1995, to include new requirements for Parties to the Convention

to provide detailed information to IMO concerning administrative measures taken to ensure compliance with the Convention. This represented the first time that IMO had been called upon to act in relation to compliance and implementation. Twenty-one years later, since 2016, the IMO Member State Audit Scheme was made mandatory in several other treaties, including SOLAS and MARPOL.

3.7 Review of Life Saving Equipment and Passenger Ship Safety

Numerous amendments have been made to SOLAS requirements, reflecting the ongoing process and work by IMO to keep safety requirements up to date as circumstances and technologies change. A new version of chapter III of SOLAS, relating to life saving appliances and arrangements, entered into force on 1 July 1998, to ensure that in the event of a catastrophe at sea, passengers and crew have the greatest chances of survival. Improved design and equipment, better fire protection, satellite communications, rescue planes and helicopters and trained personnel also contribute to improved safety at sea ("Surviving disaster—The *Titanic* and SOLAS" n.d.). In 2010, a package of SOLAS amendments adopted in 2006 entered into force, affecting passenger ships built after 1 July 2010. The amendments resulted from a comprehensive review of passenger ship safety initiated in 2000 with the aim of assessing whether the current regulations were adequate, in particular for the large passenger ships being built. Increased emphasis is placed on reducing the likelihood of accidents and improving survivability, embracing the concept of the ship "as its own best lifeboat" and a philosophy of "safe return to port" (IMO, 2006).

The scale and consistency at which these regulations and learnings can be applied varies hugely. Once again, major incidents or disasters often prompt further change. The inquiry following the sinking of the Ro-Ro ferry *Princess Victoria* in the North Sea in 1953 with the loss of 133 lives brought several safety issues to the public's attention (Watson, 2015: 231). *Princess Victoria* capsized during gale-force conditions and heavy seas that damaged its stern doors, allowing water to flood into the vehicle deck and making the ferry increasingly unstable. Efforts to control the flooding and reach safety were hampered by inadequate arrangements for draining the car deck. Only 44 of the 177 passengers and crew aboard survived. The high casualty rate was partly due to the limited availability of lifeboats and lifejackets, and the severe weather conditions which made rescue operations extremely difficult. The sinking of *Princess Victoria* had a profound impact on maritime safety regulations, particularly concerning the design and operation of roll-on/roll-off ferries. 'It also highlighted the vulnerability of this design [of vessel] to flooding, leading to stricter regulations on watertight doors, ferry design, and emergency preparedness' (Today in Celtic History, n.d.).

Ferries are still a major safety challenge in some regions, as identified by Lloyd's Register Foundation in its *Insight Report on safety in the passenger ferry industry.*

It reports that the global passenger ferry industry has averaged more than 1000 fatalities per year since the 1960s, with the great majority occurring on domestic voyages in Asia and Africa (Attwood, 2018). From 1966 to 2015, there were 750 recorded fatal accidents involving passenger vessels, resulting in 59,600 fatalities. An estimated 93% of ferry accidents occurred during domestic voyages, with 90% of fatalities occurring in just 20 countries, the highest members being in the Philippines, Bangladesh, and Indonesia. As recently as January 2026, over 51 people died with the sinking of the passenger ferry *Trisha Kerstin 3* near Basilan province, the Philippines (Marine Insight, 2026).

Lack of consistency in training and knowledge transfer, economic variances, and an inherent resistance to regulation from different industries and cultures continues to be a challenge, one that has always been there to some degree. In the fragmented shipping world that came after 1945, enforcement was yet more difficult as national flags, and even national ownership, came to have little meaning in shipping. Historian Richard Woodman observed that "The vigor of the free market often sidestepped [regulation] to leave a legacy of unsafe ships, incompetent or underpaid crews, polluted seas and declining standards." Unsurprisingly there were long delays in reaching agreement on any particular changes among member states and then in persuading sufficient national governments to adopt any agreed convention in order for it to become enforceable worldwide (Watson, 2015: 229).

The speed of technological change determined that this process had to change, and today the IMO operates a system of "tacit acceptance", enabling amendments to take effect within an agreed timescale, usually 18 to 24 months, provided objections are not received from a third of member states involved, or from member states responsible in aggregate for more than half the world's merchant fleet. (IMO, 2025). Even so, some nations are still slow in implementing and enforcing ratified conventions. Classification societies, however, are in a position to work more quickly and their role to act as Recognized Organizations to support the statutory regulator has grown in importance.

3.8 Memorandum of Understanding on Port State Control

The Hague Memorandum was developed between a number of maritime authorities in Western Europe to deal mainly with the enforcement of shipboard living and working conditions as required by International Labour Convention no. 147 (Paris MoU, n.d.). Its entry into force in 1978 was delayed by the grounding of the very large crude carrier (VLCC) *Amoco Cadiz* on rocks off the coast of Brittany, which caused terrible environmental damage. The resultant public outcry gave important new impetus to the international regulatory debate and drove maritime safety and environmental considerations up the European agenda (Djønne, 2023). As a result, a number of flag states decided to initiate a parallel system of regulation. This led to an even more stringent Memorandum of Understanding (MoU) on Port State Control (PSC) adopted by 14 European countries meeting in Paris in 1982. Principally intended to

harmonize port state control in order to eradicate sub-standard ships in their ports, each signatory agreed to inspect at least a quarter of all ships entering their ports every year. But importantly the scope of the MoU was also extended to cover maritime safety and pollution prevention.

The PSC concept began to spread as other ports became concerned that they would be seen as a safe haven for sub-standard ships. A series of agreements were reached during the 1990s, beginning with Latin America in 1992, and followed by Asia/Pacific in 1993, the Caribbean in 1996, the Indian Ocean in 1997 and the Mediterranean in 1998. With further agreements covering the Black Sea, the Gulf States, and West and South Africa, PSC now covers most of the world's shipping regions. Cooperation between the various PSC regions is intended to result in mutual acceptance of inspection results (Watson, 2015: 229). The Paris MoU is a key component of the global effort by PSC, flag states and classification societies to ensure maritime safety, compliance and environmental protection.

3.9 International Management Code for the Safe Operation of Ships and for Pollution Prevention

The importance of learning from maritime disasters was reaffirmed in 1989 when the Marine Accident Investigation Branch (MAIB) was established by the UK Government. This was in response to recommendations by the public inquiry into the *Herald of Free Enterprise* disaster that had occurred in March 1987. The ferry had set off from Zebrugge bound for Dover with its bow doors left open. Within minutes, the ship was in trouble, and in just 90s, it had capsized; 193 people died (Ships Monthly, n.d.). The subsequent criticism of management failures as a contributory cause of the disaster led the IMO to begin a process that eventually led to the 1993 adoption of the International Management Code for the Safe Operation of Ships and for Pollution Prevention (the ISM Code), which came into force in 1998.

Following the implementation of a revised Code in 2002, the PSC authorities held concerted inspections, discovering numerous breaches. This underlined the continuing need for coordination between owners, operators, flag state authorities, and classification societies to ensure masters and crew were properly trained for the safe operation of the ship and the safe delivery of cargoes (Watson, 2015: 233). "Recognizing that no two shipping companies or shipowners are the same, and that ships operate under a wide range of different conditions, the Code is based on general principles and objectives, which include assessment of all identified risks to one company's ships, personnel, and the environment and establishment of appropriate safeguards" (ISM Code, n.d.).

A key focus of the IMO is keeping legislation up to date and ensuring that it is ratified by as many countries as possible. The IMO is recognized in the 1982 United Convention on the Law of the Sea (1982 LOSC) as being the only organization competent to deal with certain international measures, such as concerning the routing

of ships. Safety of navigation (under Rule 10 of the Collision Regulations) deals with the behavior of vessels in or near traffic separation schemes adopted by the Organization. "The effectiveness of traffic separation schemes can be judged from a study made by the International Association of Institutes of Navigation (IAIN) in 1981. This showed that between 1956 and 1960 there were 60 collisions in the Strait of Dover; twenty years later, following the introduction of traffic separation schemes, this total was cut to only 16" ("Convention on the International Regulations for Preventing Collisions at Sea", n.d.). In other areas where such schemes did not exist, the number of collisions rose sharply.

For several decades, countries have worked together through the IMO to develop a comprehensive framework of international maritime conventions, with associated codes, guidelines, and recommendations ("List of IMO Conventions", n.d.). The full benefits of this framework can only be realized through ratification, effective implementation, and constant enforcement. The IMO Member State Audit Scheme (IMSAS) has reported gaps in national legislation and enforcement, indicating a need to improve regulatory effectiveness ("Member State Audit Scheme", n.d.).

The *Titanic* disaster resulted in revised domestic maritime safety laws, the first comprehensive international maritime safety agreement, and stimulated nations to establish what is now known as the IMO. It is an unanswerable question as to how many lives have been saved and accidents prevented by good design, regulations, and operating practices (Bedford & Hackett, 1996: 58). Since its adoption in 1974, the SOLAS Convention has stood as the beacon of safety standards for international shipping, ensuring the protection of lives, property, and the marine environment. Over the past five decades, the Convention has continuously evolved and expanded to address emerging challenges and keep up with technological advancements. A number of important instruments have also been adopted in more recent years. It has built a comprehensive framework which has "fostered a culture of safety consciousness among maritime nations and industry stakeholders worldwide" (IMO, 2024).

3.10 Continuing to Learn from Maritime Disaster

Apart from SOLAS, perhaps one of the most important and enduring legacies of the *Titanic* disaster is its ability to capture public imagination. Memorials, films, visualizations, museums, books, podcasts, and documentaries have made several generations aware of *Titanic*, despite it having sunk over 110 years ago. There is huge power in the stories of maritime disasters and of the people who have lost their lives in them. *Titanic* as the most famous shipwreck in the world can be seen as a powerful prompt to continue to study the numerous other incidents from which we still can learn today.

Maritime disasters and shipwrecks are examples of where safety has failed, and where lives have been lost. Learning from shipwrecks, as well as learning about what works to prevent them (such as new technologies, standard development and regulatory enforcement) is key to building and sharing the knowledge needed to continually

improve maritime safety today and for future generations. Significant shipwrecks continue to become increasingly accessible as researchers and others utilize more advanced technologies to locate and study them. The discovery of the *Titanic* in 1985 sparked relatively quick action in the US Congress that further reflected the public interest in protecting and remembering this iconic wreck.

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